



BACKEND GLITCH INC.

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PROJECT DESCRIPTION / TRAINING PLAN

Internship Title	Transport Analytics & Operations Intelligence for Wasla Platform
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Student Information

Student	Yasmine Ghomrassi
Degree Program	Master's Degree in Statistics and Data Science
University	University of Palermo (Italy)

Host Company

Company	Backend Glitch Inc.
Project	Wasla – Transport Analytics Engine
Company Supervisor	Mr. Samer Gassouma (CEO)

Internship Period

Start Date	01 July 2026
End Date	30 September 2026
Duration	3 Months (In-Person Internship)

Project Context

Backend Glitch Inc. has developed **Wasla**, a live transport management system currently deployed at Monastir Centre Station and Jammel-Monastir Station, with Ksar Hlel station planned for future deployment. The platform handles the full operational workflow: passenger bookings, vehicle dispatching (louages/shared taxis), day pass management,

route scheduling, ticket printing, staff shift tracking, and real-time station monitoring with a fleet of 113 active vehicles serving 5 destination routes.

The platform has been accumulating operational data since April 2026, covering passenger bookings, vehicle trips, revenue, fleet utilisation, and station operations. This data represents a valuable resource for business intelligence and operational optimisation.

This internship focuses on the application of **Statistics**, **Data Science**, and **Cloud Computing** techniques to analyse operational transport data, develop passenger flow forecasting models, create anomaly detection systems, build analytical dashboards, and deploy cloud-based analytical solutions.

Project Objectives

The main objectives of the internship are:

1. Understand the architecture and operational workflow of the Wasla platform.
2. Analyse passenger booking data, revenue streams, and fleet utilisation using statistical techniques.
3. Develop time series forecasting models for passenger volume prediction.
4. Design and implement anomaly detection systems for booking pattern monitoring.
5. Create analytical dashboards for real-time operations monitoring and KPI tracking.
6. Deploy analytical services within a cloud environment (Docker, Linux VPS).
7. Gain practical experience in data-driven decision-making systems for transport operations.

Detailed Timeline and Activities

Phase 1 – Onboarding, Observation and Learning

Period: 01 July 2026 – 14 July 2026

Activities

- Introduction to Backend Glitch Inc. and its products.
- Familiarization with the Wasla platform and station operations workflow.
- Meetings with technical and operations teams.
- Understanding available datasets and system architecture.
- Study of existing infrastructure and cloud environment.
- Definition of project goals and KPIs.

Deliverables

- Learning and observation report.
- Project roadmap.
- KPI definition document.

Phase 2 – Data Collection and Preparation

Period: 15 July 2026 – 31 July 2026

Activities

- Connection to the production PostgreSQL database.
- Extraction of booking, trip, vehicle, and station data.
- Data cleaning and preprocessing.
- Handling missing and inconsistent data.
- Feature engineering (time-based features, seasonality markers).
- Initial exploratory data analysis.

Deliverables

- Clean analytical dataset.
- Data preparation report.

Phase 3 – Statistical Analysis

Period: 01 August 2026 – 15 August 2026

Activities

- Descriptive statistical analysis of booking volumes and revenue.
- Passenger flow pattern analysis (hourly, daily, seasonal).
- Route performance comparison across 5 destinations.
- Fleet utilisation and load factor analysis.
- Peak hour identification and trend analysis.
- Correlation analysis between operational indicators.

Deliverables

- Statistical analysis report.
- KPI analysis report.
- Data visualization outputs.

Phase 4 – Data Science and Predictive Analytics

Period: 16 August 2026 – 05 September 2026

Activities

- Feature selection and engineering for forecasting models.
- Development of time series forecasting models (ARIMA, Prophet).
- Passenger volume prediction (7-day and 30-day forecasts).
- Anomaly detection model development (statistical thresholds, Isolation Forest).

- Model validation and performance evaluation (MAPE, RMSE, precision/recall).
- Alert threshold calibration and severity classification.

Techniques

- ARIMA / SARIMA.
- Facebook Prophet.
- Isolation Forest.
- Statistical Process Control.

Deliverables

- Passenger volume forecasting model.
- Anomaly detection system.
- Model evaluation report.

Phase 5 – Dashboard Development and Cloud Deployment

Period: 06 September 2026 – 20 September 2026

Activities

- Design and implementation of analytical dashboards (Next.js).
- Visualization of passenger flow, revenue, fleet, and forecast KPIs.
- Database integration (PostgreSQL read-only connection).
- Containerisation of analytics API (FastAPI) and dashboard (Docker).
- Cloud deployment to production Linux VPS with Nginx reverse proxy.
- Monitoring and testing.

Deliverables

- Interactive analytics dashboard.
- Cloud deployment documentation.

Phase 6 – Validation and Final Reporting

Period: 21 September 2026 – 30 September 2026

Activities

- Validation of analytical results and model performance.
- Documentation of methodologies and findings.
- Preparation of final presentation.

Deliverables

- Final internship report.
- Technical documentation.
- Final presentation.

Expected Outcomes

At the end of the internship, the student is expected to have:

- Applied statistical analysis techniques to real-world transport datasets.
- Developed passenger flow forecasting and anomaly detection models.
- Created analytical dashboards for operations monitoring and reporting.
- Acquired practical experience in cloud-based data solutions and deployment.
- Produced a complete technical and scientific report documenting the work performed.

Signatures

Student

Name Yasmine Ghomrassi



Signature

Date 13 June 2026

Host Company Supervisor

Name Mr. Samer Gassouma
Chief Executive Officer (CEO)
Backend Glitch Inc.

Signature



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Date 13 June 2026

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